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Minor Project Proposal using JDBC

Security Incident&Asset Tracker

1. Project Title

Security Incident & Asset Tracker — A JDBC-Based Application for Monitoring IT Assets and Security Incidents

2. Introduction

I'm currently working toward a career in cybersecurity, starting with an entry-level SOC Analyst role and eventually moving into cloud security. While learning the fundamentals of how SOC teams operate, one thing stood out to me — a huge part of the job isn't just detecting threats, it's keeping track of what you own and what's happening to it. Every asset an organization has (laptops, servers, routers) is a potential target, and every incident reported against those assets needs to be logged, prioritized, and resolved without anything slipping through the cracks.

For this assignment, instead of picking a generic problem like a library or student database, I wanted to build something closer to the kind of tool a SOC team would actually use day to day. This project, the Security Incident & Asset Tracker, is a Java application built using JDBC that connects to a MariaDB database to manage an organization's IT assets and the security incidents reported against them. It's a small-scale version of the asset and incident management systems used in real security operations — built here to practice JDBC concepts while also giving me something relevant to talk about when I eventually interview for SOC roles.

3. Problem Statement

Manually tracking IT assets and security incidents — through spreadsheets or paper logs — doesn't scale and makes it hard to see the full picture: which asset had which incident, how severe it was, and whether it's still open. There's a need for a simple, centralized system where assets and incidents are properly linked, incidents can be classified by severity, and open high-priority issues can be found quickly instead of getting buried.

4. Objectives

Design a relational database to store asset and incident records.

Build a Java application using JDBC to handle Create, Read, Update, and Delete (CRUD) operations.

Let users add, view, update, and delete asset records.

Let users report new incidents and link each one to a specific asset.

Classify incidents by severity (Low, Medium, High, Critical) and track status (Open/Resolved).

Add a way to filter and view only open incidents of a chosen severity, since in a real SOC, knowing what's urgent matters more than knowing everything at once.

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5. Scope of the Project

This is a single-user, console-based application built for academic purposes — not a production security tool. It handles asset inventory and incident tracking for one organization. It doesn't include things like user login/role-based access, real-time monitoring, or automated threat detection — those are well beyond the scope of a minor project, though they're the kind of features a full version could eventually grow into. The interface is console-based for now; a GUI could be a natural next step if I extend this later.

6. Functional Requirements

- Add a new asset with name, type, owner, and location.
- View a list of all assets.
- Update the owner of an existing asset.
- Delete an asset record.
- Report a new incident linked to an asset, with a description and severity level.
- View all incidents.
- Filter and view only Open incidents of a chosen severity.
- Mark an incident as Resolved, recording the date it was resolved.
- Delete an incident record.

7. Non-Functional Requirements

- Reliability: Data should persist correctly in the database with no loss.
- Usability: The console menu should be simple enough that anyone can use it without instructions.
- Performance: Operations should respond quickly for a single-user application — no noticeable lag.
- Maintainability: Code is organized using the DAO (Data Access Object) pattern, so database logic is separated from the menu/interface logic. This made debugging a lot easier while building it.
- Security: Every database query uses PreparedStatement instead of plain string concatenation, to avoid SQL injection — a habit worth building early, given where I'm headed career-wise.

8. Software and Hardware Requirements

Software Requirements

- Operating System: Kali Linux
- Programming Language: Java (JDK 17 or later)
- IDE: Apache NetBeans
- Database: MariaDB
- JDBC Driver: MariaDB Connector/J (mariadb-java-client.jar)

Hardware Requirements

- Processor: Intel Core i5 or equivalent
- RAM: 4 GB minimum (8 GB or more recommended)
- Storage: 500 MB free disk space for development environment and database

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9. Database Design

The database SecurityDB has two related tables: Assets and Incidents. Every incident is linked back to a specific asset through a foreign key, so nothing is ever tracked in isolation — the same way real asset management works.

Table:Assets

Field Name	Data Type	Key	Null Able	Description
asset_id	INT(Auto Increment)	Primary Key	No	Unique identifier for each asset
asset_name	VARCHAR	-	No	Name of the asset
asset_type	VARCHAR	-	Yes	Type eg.Laptop,Server, Router
owner	VARCHAR	-	Yes	Person/Department responsible
location	VARCHAR	-	Yes	Physical or network location

Table:Incident

Field Name	Data Type	Key	Null Able	Description
incident_id	INT(Auto Increment)	Primary Key	No	Unique identifier for each asset
asset_id	VARCHAR	Foreign Key	Yes	References Assets(asset_id)
description	VARCHAR	-	Yes	Description of the incident
severnity	VARCHAR	-	Yes	Low / Medium / High / Crirical
status	VARCHAR	-	No (default 'open')	Open or Resolved
reported_date	DATE	-	Yes	Date insident was reported
resolved_date	DATE	-	Yes	Date incident was resolved

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Database setup(MariaDB Terminal)

- **Assets table created**
- **Incident table created**
- **Describe output**

```

sudo mysql -u root -p

Session  Actions  Edit  View  Help
+-----+-----+-----+-----+
| Field | Type | Null | K |
+-----+-----+-----+-----+
| asset_id | int(11) | NO | P |
| asset_name | varchar(100) | NO | |
| asset_type | varchar(50) | YES | |
| owner | varchar(100) | YES | |
| location | varchar(100) | YES | |
+-----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [SecurityDB]> describe Assets;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| asset_id | int(11) | NO | PRI | NULL | auto_increment |
| asset_name | varchar(100) | NO | | NULL | |
| asset_type | varchar(50) | YES | | NULL | |
| owner | varchar(100) | YES | | NULL | |
| location | varchar(100) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [SecurityDB]> describe Incidents;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| incident_id | int(11) | NO | PRI | NULL | auto_increment |
| asset_id | int(11) | YES | MUL | NULL | |
| description | varchar(255) | YES | | NULL | |
| severity | varchar(20) | YES | | NULL | |
| status | varchar(20) | YES | | Open | |
| reported_date | date | YES | | NULL | |
| resolved_date | date | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.001 sec)

MariaDB [SecurityDB]>

```

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10. Expected Output

When the application runs, it shows a console menu for managing assets and incidents. Expected outputs include:

A "Connected successfully" message on startup.

Confirmation after adding, updating, or deleting an asset (e.g., "Asset added successfully").

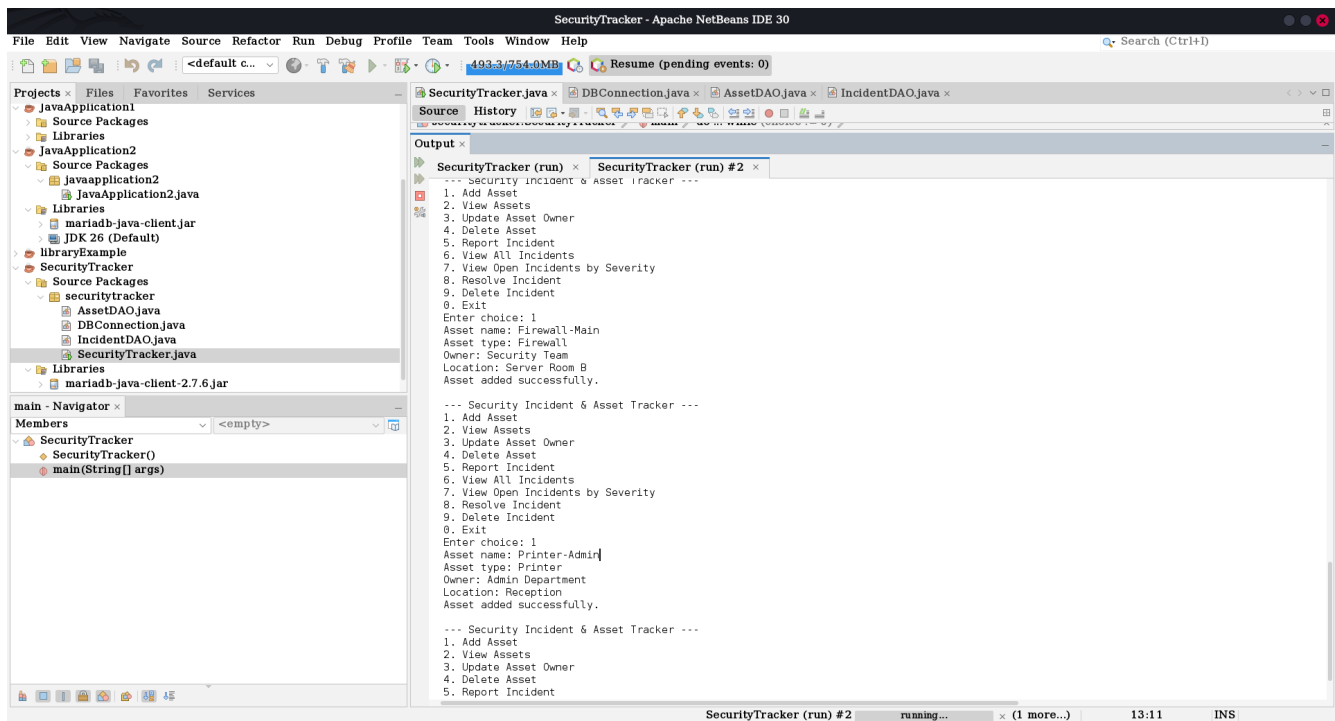
A formatted list of all assets.

Confirmation after reporting a new incident.

A formatted list of all incidents, showing severity and status.

A filtered list of only Open incidents for a chosen severity.

Confirmation and updated status when an incident is marked Resolved.



```
SecurityTracker - Apache NetBeans IDE 3.0
File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help
493.3/754.0MB Resume (pending events: 0)

Projects | Files | Favorites | Services
JavaApplication1
  Source Packages
  Libraries
JavaApplication2
  Source Packages
  JavaApplication2.java
  Libraries
    mariadb-java-client.jar
    JDK 26 (Default)
libraryExample
SecurityTracker
  Source Packages
  securitytracker
    AssetDAO.java
    DBConnection.java
    IncidentDAO.java
    SecurityTracker.java
  Libraries
    mariadb-java-client-2.7.6.jar

main - Navigator
Members
  SecurityTracker
    SecurityTracker()
    main(String[] args)

Source
SecurityTracker.java
DBConnection.java
AssetDAO.java
IncidentDAO.java

Output
SecurityTracker (run) #2
--- Security Incident & Asset Tracker ---
1. Add Asset
2. View Assets
3. Update Asset Owner
4. Delete Asset
5. Report Incident
6. View All Incidents
7. View Open Incidents by Severity
8. Resolve Incident
9. Delete Incident
0. Exit
Enter choice: 1
Asset name: Firewall-Main
Asset type: Firewall
Owner: Security Team
Location: Server Room B
Asset added successfully.

--- Security Incident & Asset Tracker ---
1. Add Asset
2. View Assets
3. Update Asset Owner
4. Delete Asset
5. Report Incident
6. View All Incidents
7. View Open Incidents by Severity
8. Resolve Incident
9. Delete Incident
0. Exit
Enter choice: 1
Asset name: Printer-Admin
Asset type: Printer
Owner: Admin Department
Location: Reception
Asset added successfully.

--- Security Incident & Asset Tracker ---
1. Add Asset
2. View Assets
3. Update Asset Owner
4. Delete Asset
5. Report Incident
```

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```
SecurityTracker (run) x SecurityTracker (run) #2 x
7. View Open Incidents by Severity
8. Resolve Incident
9. Delete Incident
0. Exit
Enter choice: 1
Asset name: Printer-Admin
Asset type: Printer
Owner: Admin Department
Location: Reception
Asset added successfully.

--- Security Incident & Asset Tracker ---
1. Add Asset
2. View Assets
3. Update Asset Owner
4. Delete Asset
5. Report Incident
6. View All Incidents
7. View Open Incidents by Severity
8. Resolve Incident
9. Delete Incident
0. Exit
Enter choice: 2
1 | Laptop | Laptop | IT Department | Office
3 | Server-801 | Server | IT Department | Server Room C6
4 | Core-Router-82 | Router | Network Team | Data Center
5 | Desktop-HR-07 | Desktop | HR Department | HR Office, 2nd Floor
6 | Firewall-Main | Firewall | Security Team | Server Room B
7 | Printer-Admin | Printer | Admin Department | Reception

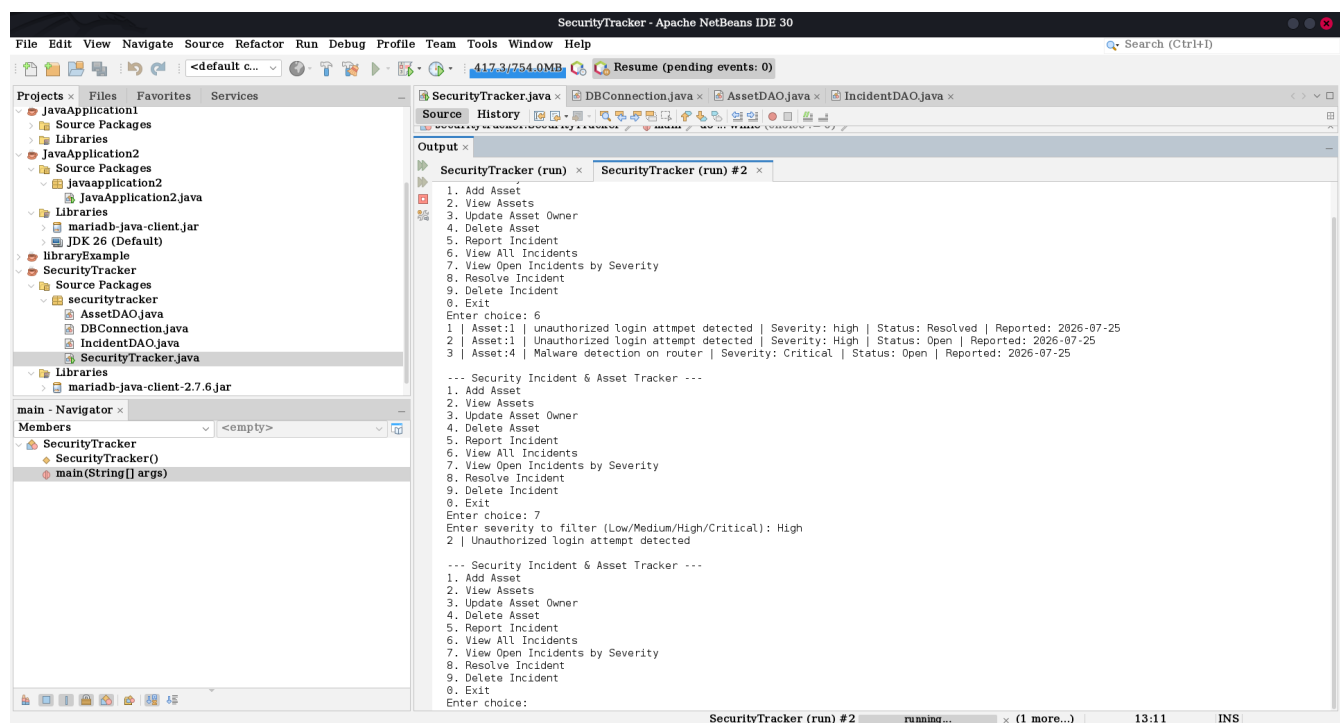
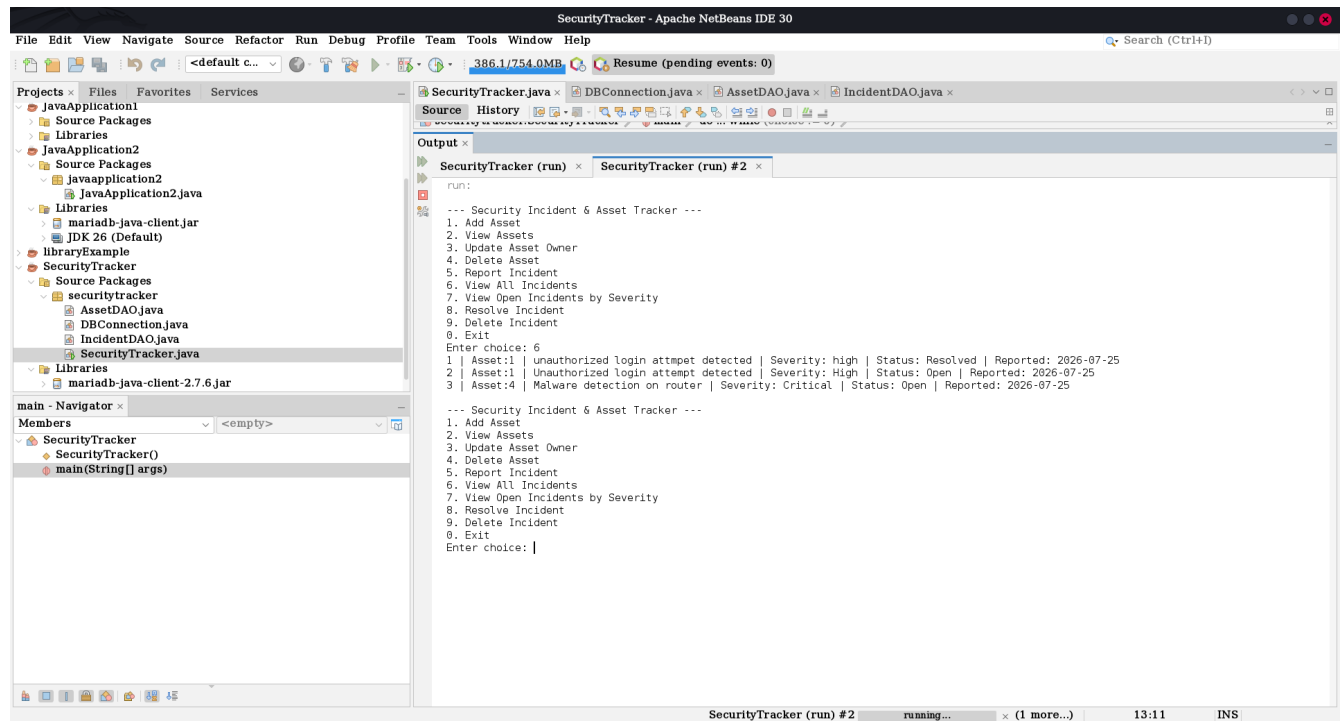
--- Security Incident & Asset Tracker ---
1. Add Asset
2. View Assets
3. Update Asset Owner
4. Delete Asset
5. Report Incident
6. View All Incidents
7. View Open Incidents by Severity
8. Resolve Incident
9. Delete Incident
0. Exit
Enter choice: |
```

```
SecurityTracker (run) x SecurityTracker (run) #2 x
6. View All Incidents
7. View Open Incidents by Severity
8. Resolve Incident
9. Delete Incident
0. Exit
Enter choice: 2
1 | Laptop | Laptop | IT Department | Office
3 | Server-801 | Server | IT Department | Server Room C6
4 | Core-Router-82 | Router | Network Team | Data Center
5 | Desktop-HR-07 | Desktop | HR Department | HR Office, 2nd Floor
6 | Firewall-Main | Firewall | Security Team | Server Room B
7 | Printer-Admin | Printer | Admin Department | Reception

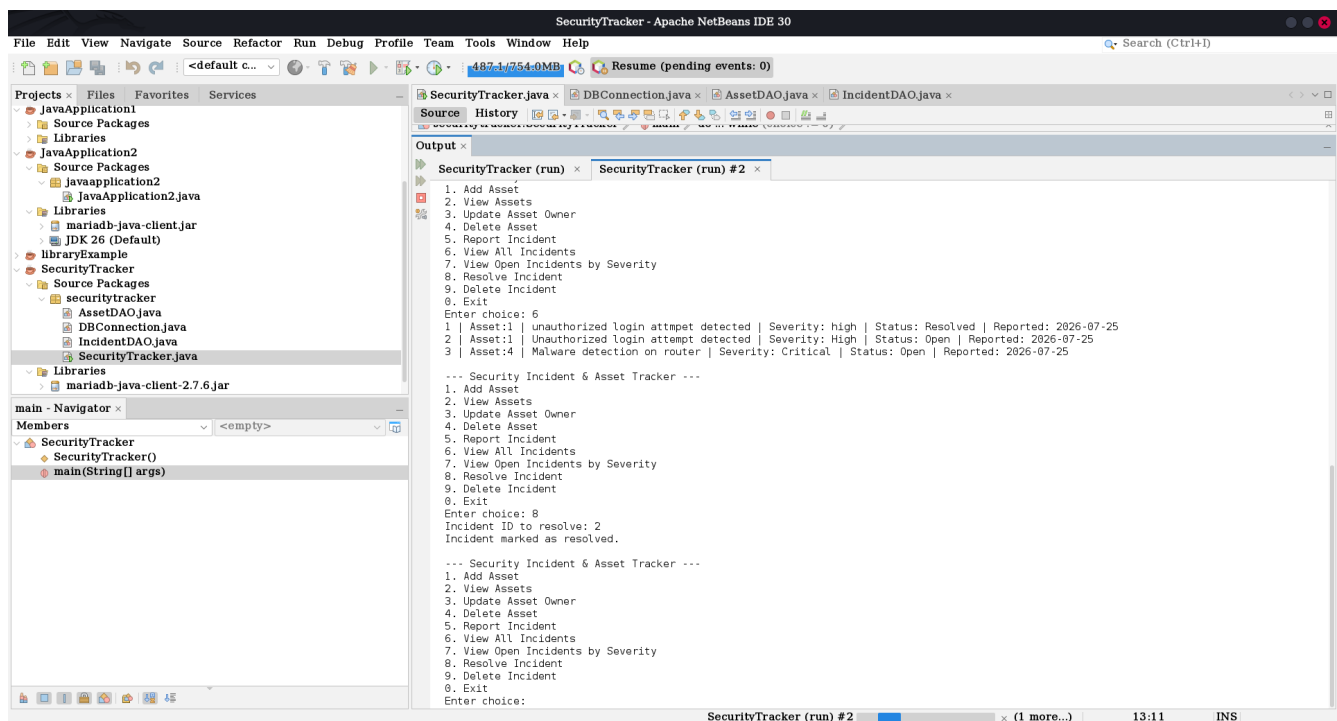
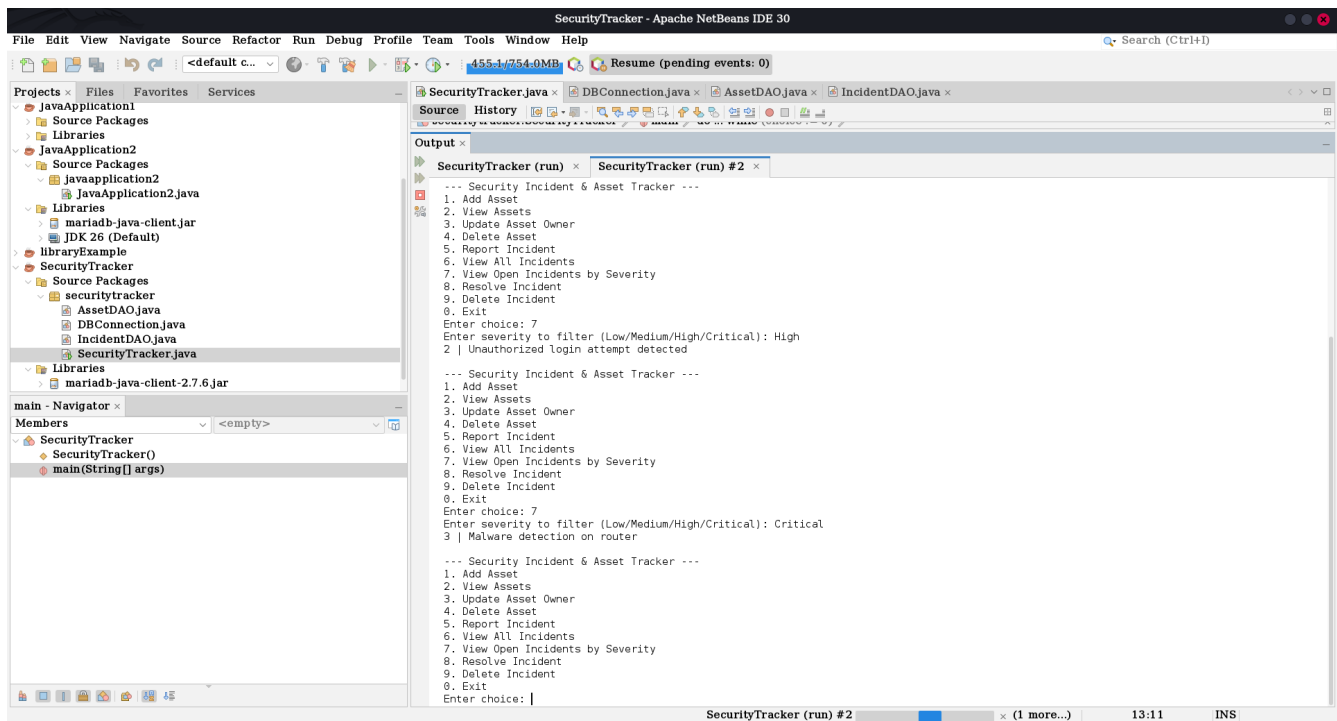
--- Security Incident & Asset Tracker ---
1. Add Asset
2. View Assets
3. Update Asset Owner
4. Delete Asset
5. Report Incident
6. View All Incidents
7. View Open Incidents by Severity
8. Resolve Incident
9. Delete Incident
0. Exit
Enter choice: 5
Asset ID this incident belongs to: 1
Description: Unauthorized login attempt detected
Severity (Low/Medium/High/Critical): High
Incident reported successfully.

--- Security Incident & Asset Tracker ---
1. Add Asset
2. View Assets
3. Update Asset Owner
4. Delete Asset
5. Report Incident
6. View All Incidents
7. View Open Incidents by Severity
8. Resolve Incident
9. Delete Incident
0. Exit
Enter choice:
```

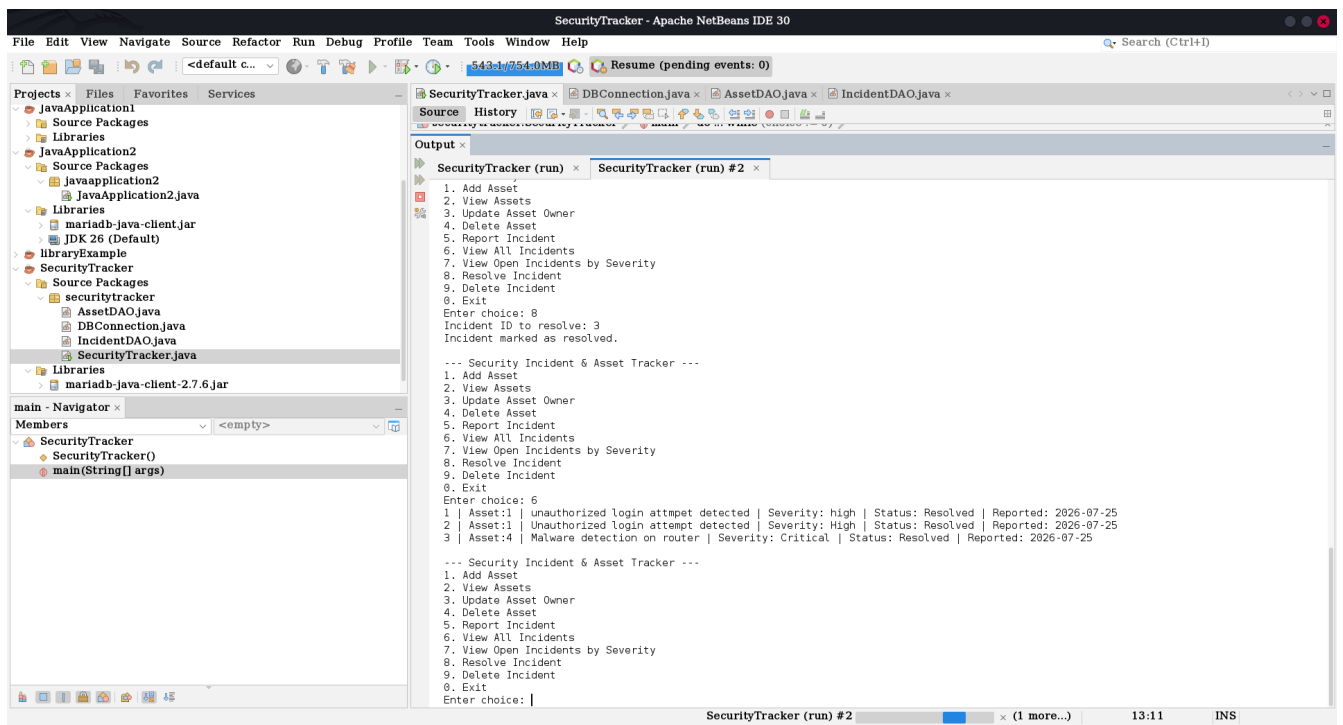
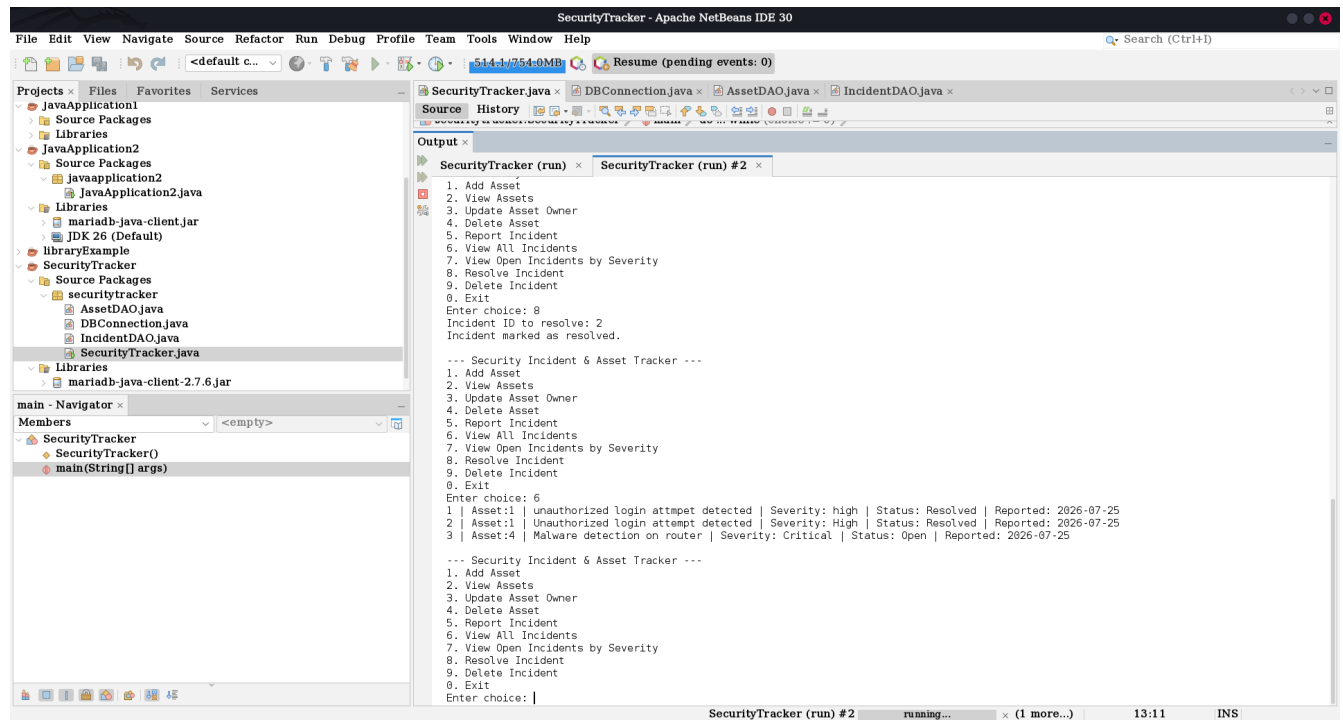
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11. References

Oracle Corporation, "Java SE JDBC Documentation", <https://docs.oracle.com/javase/tutorial/jdbc/>
MariaDB Foundation, "MariaDB Connector/J Documentation", <https://mariadb.com/kb/en/mariadb-connector-j/>

Apache NetBeans Documentation, <https://netbeans.apache.org/>

Course materials and lab exercises, J2EE / JDBC coursework, St. Aloysius University

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J2EE ASSIGNMENT

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